1. Identity

Document	STAR-PKG-STAR-MRP-EN · v2.1 · 2026-11-01
Package	star-mrp
Version	0.1.0
Capability	star.attitude.mrp
Maturity	RELEASED
Python	⩾3.9
Source commit	94fcf7c
Source	https://github.com/Andrea07072000/S.T.A.R.-/tree/main/packages/star_mrp

2. Licence and obligations

Apache License 2.0. You may use, copy, modify and distribute it, also commercially, and you receive a patent licence from the contributors. When you redistribute it: keep the LICENSE file and the copyright notices (and a NOTICE file, if a future version adds one), and state in modified files that you changed them. You may not use the S.T.A.R. name to endorse your product.

3. What it needs at run time

Nothing beyond the Python standard library: it works offline.

4. Integrity

SHA-256 fingerprints of the files published on this website. Check a downloaded file with:

```
python -c "import hashlib,sys;print(hashlib.sha256(open(sys.argv[1],'rb').read()).hexdigest())" star_mrp-0.1.0-py3-none-any.whl
```

FILE	SHA-256
star_mrp-0.1.0-py3-none-any.whl	f3bfb4e876ab9c5179b8023a9902b087d99db35f4796ed6c1e316487bf1a21ae
star_mrp.py	dd6d256d9a3a171e89e377271cc6e8d48ecb22661109db2db20453cefd25af04
star_mrp-0.1.0-source.zip	a9626df085ee89b756530d8fe59e372fc855bee91229a3a50d2fdbbeb716c884

5. What it does, and its limits

Modified Rodrigues parameters: conversion with quaternions, direction cosine matrices and rotation vectors, shadow set and switching, composition of rotations through quaternions (no division by zero at a full turn) and the kinematic equation; within 6e-16 of Basilisk and 2e-15 of SciPy on 601 pairs with norms from 1e-8 to 5.

Kinematics only: no dynamics, no control law, no integrator (`mrp_rate` is the right-hand side; integrating it and switching to the shadow set is left to the caller). At exactly half a turn the two MRPs of norm 1 are equally short and either may be returned. No Gibbs vector (classical Rodrigues parameters), no Euler angles (see star_euler). No published numerical table is reproduced: the references are cited for the formulas.

- `mrp_from_quaternion`, `quaternion_from_mrp`, `dcm_from_mrp`, `rotation_vector_from_mrp`, `mrp_from_rotation_vector` convert between sigma = e tan(phi/4) and the other representations; `shadow` and `switch` give the other MRP of the same attitude; `compose(first, second)` chains two rotations; `mrp_rate(s, omega)` is the kinematic equation.
- Quaternions are scalar first; the matrix takes reference to body; results of conversions and of `compose` are on the short rotation (|sigma| <= 1, quaternion with w >= 0, angle in [0, pi]). `compose` goes through quaternions, so two rotations that add up to a full turn give zero and not a division by zero.
- Measured on 601 pairs of MRPs with norms from 1e-8 to 5 (a third beyond 1): eight operations agree with Basilisk within 6e-16 and seven with SciPy within 2e-15.
- Every function returns floats or raises `ValueError`: vectors that are not three (or four) finite real numbers within 1e150, a zero quaternion, the shadow of the zero MRP, a threshold that is not positive.

6. Export, use and certification

The package is published and freely available. That usually takes software out of export-control lists, but not every use of it: read the page on space activities and export control before using it in a controlled programme. It is not certified or qualified for flight; the evidence describes what was tested and how.

7. Citation and bill of materials

```
@software{cavazzini_star_mrp_0_1_0,
  author  = {Cavazzini, Andrea},
  title   = {{star-mrp}: Modified Rodrigues parameters},
  version = {0.1.0},
  year    = {2026},
  url     = {https://andrea07072000.github.io/S.T.A.R.-/software/star_mrp/},
  note    = {Apache-2.0. Source: https://github.com/Andrea07072000/S.T.A.R.-, commit 94fcf7c}
}
```

CITATION.cff · BibTeX · SBOM CycloneDX 1.5 · SBOM SPDX 2.3

8. Warranty and liability

Provided "as is", without warranties or conditions of any kind (Apache License 2.0, sections 7 and 8), to the extent the law allows. This never excludes liability that the law does not allow to exclude.

Related: space activities and export control · acceptable use · evidence and limits · licence texts